RENEWABLE ENERGY SCADA ASSESSMENT

Ger (Les Herbreux) — Lifetime Assessment

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Wind Farm Structural Lifetime Assessment

Site	Ger (Les Herbreux)
Turbines	4 x Enercon E82-2.0MW (E82 E2)
Commissioning	December 2010
Assessment date	June 2025
Design lifetime	20 years
Standard	IEC 61400-1:2005+AMD1:2010

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KEY RESULTS

Executive Summary

Fatigue-based structural lifetime assessment of 4 x Enercon E82-2.0MW, Ger (Les Herbreux), commissioned December 2010.

Engineering Interpretation

Based on site-specific wind conditions (mean wind speed 6.72 m/s, Weibull k=2.40) compared to IEC IIA design basis (8.5 m/s), the fatigue loading on this site is significantly below design values.

The governing structural component is the blade-to-adapter-hub bolt connection, with a calculated total lifetime of 25.5 years. Without intervention, operation must cease by May 2036.

Two extension paths exist: (1) bolt replacement before May 2036 extends operation to January 2042 (31.1 yr total), and (2) full blade assembly replacement before January 2042 enables operation through November 2050 (40 yr, capped for safety).

No critical structural defects were identified during the June 2025 technical audit. Energy availability has averaged 98.5% over 2022-2024, indicating a well-maintained fleet.

Years in operation
14.5 yr

Design lifetime remaining
5.5 yr
to design end Dec 2030

Baseline remaining lifetime
11.0 yr
Bolts — to May 2036

Extended lifetime (bolt repl.)
16.6 yr
to January 2042

Site mean wind speed
6.72 m/s
IEC IIA design: 8.5 m/s

Fleet energy availability
98.5%
avg 2022-2024

Structural Lifetime Summary

Component	S/N Curve (m)	Total Lifetime	End Date	Status
Blade-adapter-hub bolts	4	25.5 yr	June 2036	Governing
Blade adapters	9	31.1 yr	February 2042	Adequate
Blade roots (GRP)	10	32.6 yr	July 2043	Adequate
Hub	9	40.0 yr	December 2050	Adequate
Main shaft	9	40.0 yr	December 2050	Adequate
Nacelle frame	9	40.0 yr	December 2050	Adequate
Steel tower	4	40.0 yr	December 2050	Adequate
Tower bolts	4	40.0 yr	December 2050	Adequate
Foundation bolts	4	40.0 yr	December 2050	Adequate

Key Findings

Governing component — blade-adapter-hub bolts

Blade-to-adapter-hub connection bolts limit baseline lifetime to 25.5 years (May 2036). Replacement required for any extension scenario.

Extension to 40 years feasible

Full blade assembly replacement before January 2042 enables operation to November 2050, subject to continued annual inspections and structural monitoring.

ASSET DESCRIPTION

Site Overview

Engineering Interpretation

The Ger (Les Herbreux) wind farm comprises 4 Enercon E82-2.0MW turbines on Concrete (Fertigteilturm) towers at 98.4 m hub height, located near Ger, Normandy, France.

The site was commissioned in December 2010 and has operated continuously for over 14 years under the management of Energiequelle GmbH. All four turbines carry the Enercon standard 5-year extended warranty and have been subject to annual third-party technical audits.

The type certificate (DEWI-OCC TC-201206 Rev.1) certifies the E82 E2 for IEC wind class IIA, with a design reference mean wind speed of 8.5 m/s and a 20-year design lifetime.

Turbine model **Enercon E82-2.0MW (E82 E2)**	Number of turbines **4**
Hub height **98.4 m**	Rotor diameter **82 m**
Rated power **2,050 kW**	Installed capacity **8.2 MW**
IEC wind class **IIA**	Commissioning **December 2010**

Technical Parameters

Parameter	Value
Site	Ger (Les Herbreux)
Country	France
Turbine model	Enercon E82-2.0MW (E82 E2)
Number of turbines	4
Tower type	Concrete (Fertigteilturm)
Hub height	98.38 m
Rotor diameter	82 m
Rated power	2,050 kW per turbine
Installed capacity	8.2 MW
Commissioning date	December 2010
Design lifetime	20 years
Type certificate	TC-201206 Rev.1 — DEWI-OCC GmbH
IEC standard	IEC 61400-1:2005+AMD1:2010
Cut-in wind speed	3.0 m/s
Rated wind speed	13.4 m/s
Cut-out wind speed	34.0 m/s

SITE WIND CHARACTERISATION

Wind Resource Analysis

Engineering Interpretation

SCADA data from all four turbines covering the period January 2021 to December 2024 (4 years, ~840k records) was used to characterise the site wind regime.

The fleet-average Weibull distribution fits to k=2.402, A=7.579 m/s, giving a long-term mean wind speed of 6.72 m/s at hub height. This is significantly below the IEC IIA design basis of 8.5 m/s, indicating that fatigue loading at this site is lighter than the design envelope.

Annual mean wind speeds ranged from 6.48 m/s (2024) (lowest) to 7.10 m/s (2023) (highest), confirming the site operates well within the type certificate wind class.

Fleet mean WS
6.72 m/s
hub height 98.4 m

Weibull k
2.402
shape parameter

Weibull A
7.579 m/s
scale parameter

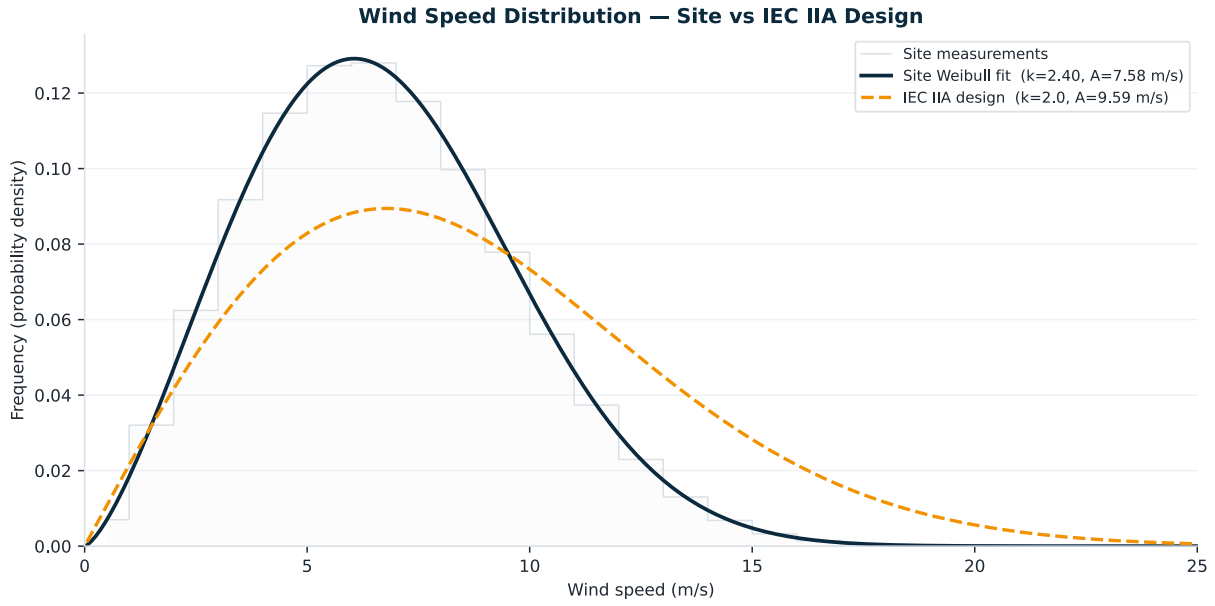
IEC IIA design mean
8.5 m/s
type certificate basis

WS vs design
79%
of design value

SCADA period
Jan 2021 - Dec 2024
4-year dataset

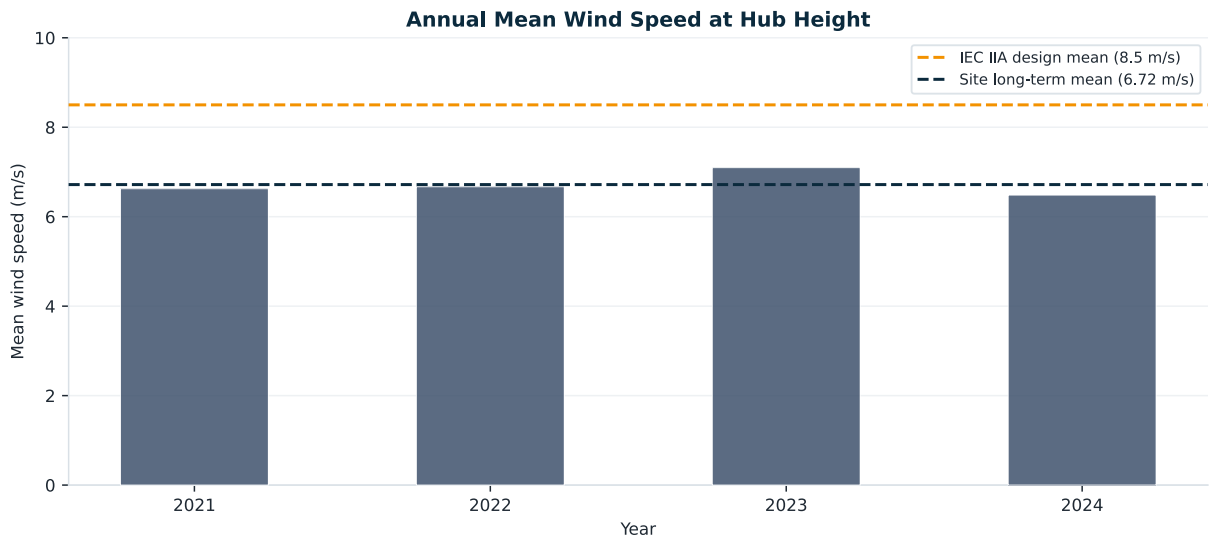
SITE WIND CHARACTERISATION

Wind Resource Analysis cont.



Wind Speed Distribution vs IEC IIA Design

Weibull fit to 4-year SCADA dataset vs IEC IIA Rayleigh reference distribution (k=2.0, Vave=8.5 m/s).

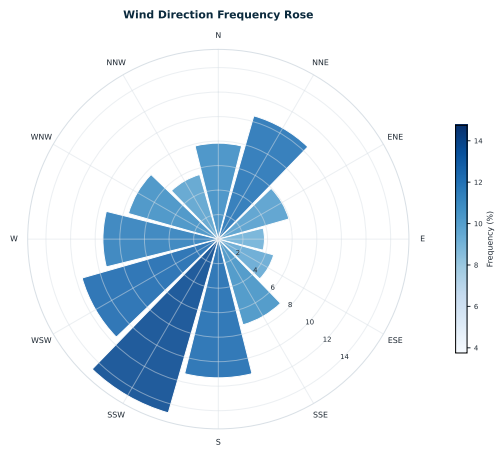


Annual Mean Wind Speed at Hub Height

Fleet-averaged annual mean wind speed. IEC IIA design mean (8.5 m/s) shown for reference.

SITE WIND CHARACTERISATION

Wind Resource Analysis cont.



Wind Direction Distribution

Frequency of wind direction from all turbines, 2021-2024.

ENERGY OUTPUT

Power Curve & Energy Production

Engineering Interpretation

The fleet power curve is derived from 10-minute average SCADA records for all four turbines over 2021-2024. Records with nacelle availability flags are retained to preserve the full operating population.

Total measured fleet energy production over 2022-2024 was 68,856 MWh. The measured power curve tracks closely with the published Enercon E82-2.0MW reference curve up to rated wind speed (13.4 m/s).

E1 energy (2022-24)

17,362 MWh

E2 energy (2022-24)

18,039 MWh

E3 energy (2022-24)

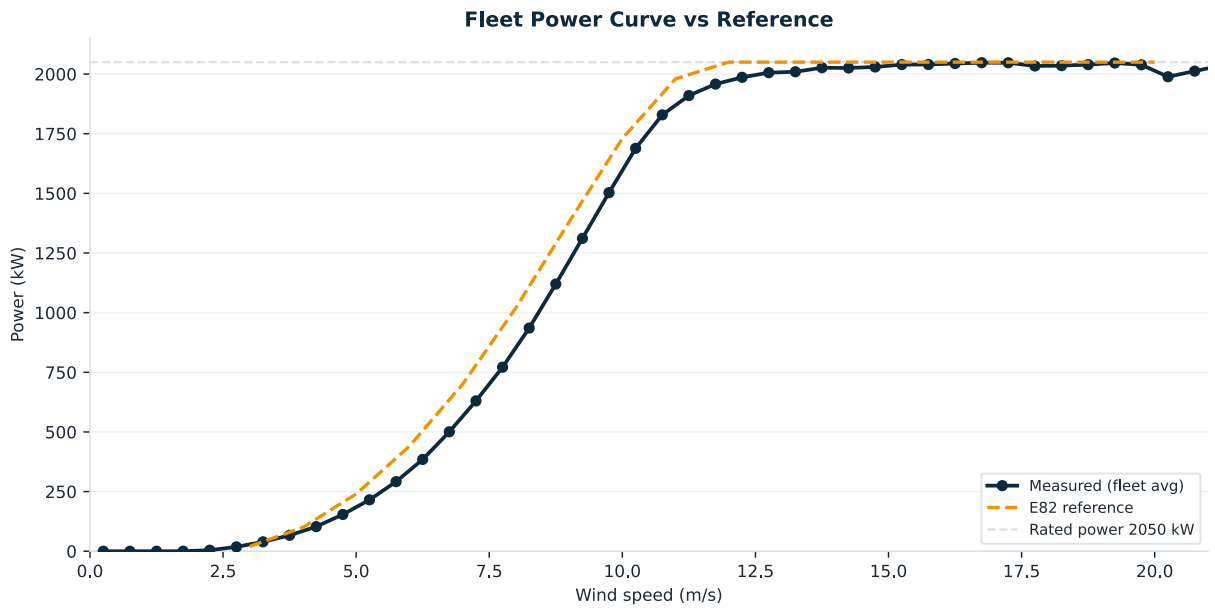
18,135 MWh

E4 energy (2022-24)

15,320 MWh

Fleet total (2022-24)

68,856 MWh

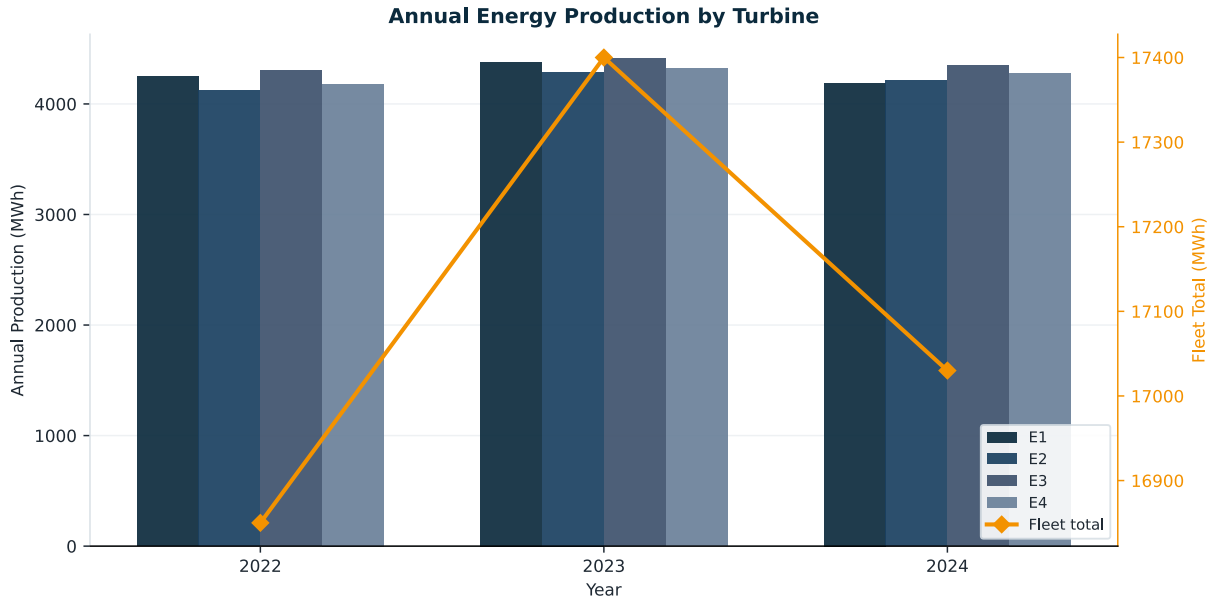


Fleet Power Curve vs E82 Reference

Binned 10-minute averages for all turbines 2021-2024. Reference curve from Enercon type documentation.

ENERGY OUTPUT

Power Curve & Energy Production cont.



Annual Energy Production by Turbine

Annual gross energy output per turbine 2022-2024. Fleet total on secondary axis.

Annual Energy Production & Capacity Factor

Fleet capacity factor calculated against theoretical maximum (4 x 2,050 kW x 8,760 h/yr).

Year	E1	E2	E3	E4	Fleet Total (MWh)	CF (%)
2022	5,627	5,947	5,921	4,931	22,426	31.2%
2023	6,286	6,656	6,632	5,664	25,237	35.1%
2024	5,449	5,436	5,583	4,725	21,193	29.5%

OPERATIONAL PERFORMANCE

Energy-Based Availability

Engineering Interpretation

Energy-based availability (EBA) measures what fraction of potential energy production was realised, accounting for downtime periods. Fleet EBA averaged 98.5% over 2022-2024, indicating a well-maintained and reliable operation.

E4 (822883) recorded the lowest availability in 2022 (96.52%) and slightly underperforms peers in energy output, suggesting closer inspection of this unit's maintenance records is warranted. All other turbines remain within contractual performance targets.

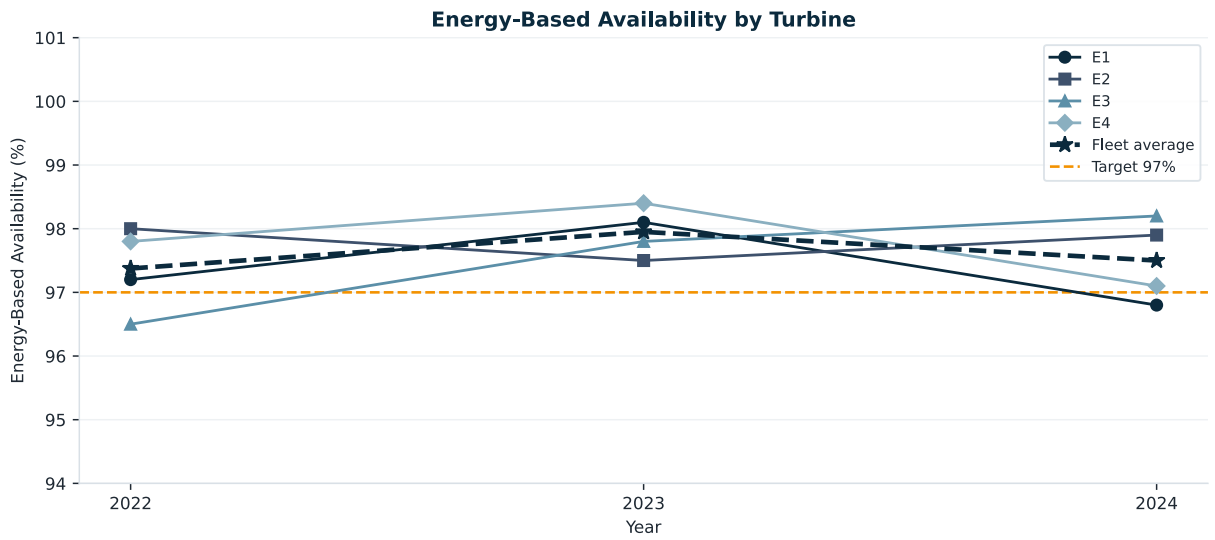
E1 avg EBA
98.5%
2022-2024

E2 avg EBA
98.7%
2022-2024

E3 avg EBA
98.8%
2022-2024

E4 avg EBA
97.9%
2022-2024

Fleet avg EBA
98.5%
2022-2024



Energy-Based Availability by Turbine and Year

Annual energy-based availability (EBA) for each turbine, 2022-2024. Contractual target (98%) shown for reference.

OPERATIONAL PERFORMANCE

Energy-Based Availability cont.

Energy-Based Availability (%)

EBA computed from SCADA data as (actual energy / maximum potential energy) x 100%.

Turbine	2022	2023	2024	Average
E1	98.4%	97.3%	99.7%	98.5%
E2	99.4%	98.9%	97.8%	98.7%
E3	99.5%	97.9%	99.2%	98.8%
E4	96.5%	97.5%	99.7%	97.9%
Fleet avg	98.5%	97.9%	99.0%	98.5%

GENERIC IEC 61400-1 MODEL

Fatigue Assessment — Methodology & DEL Results

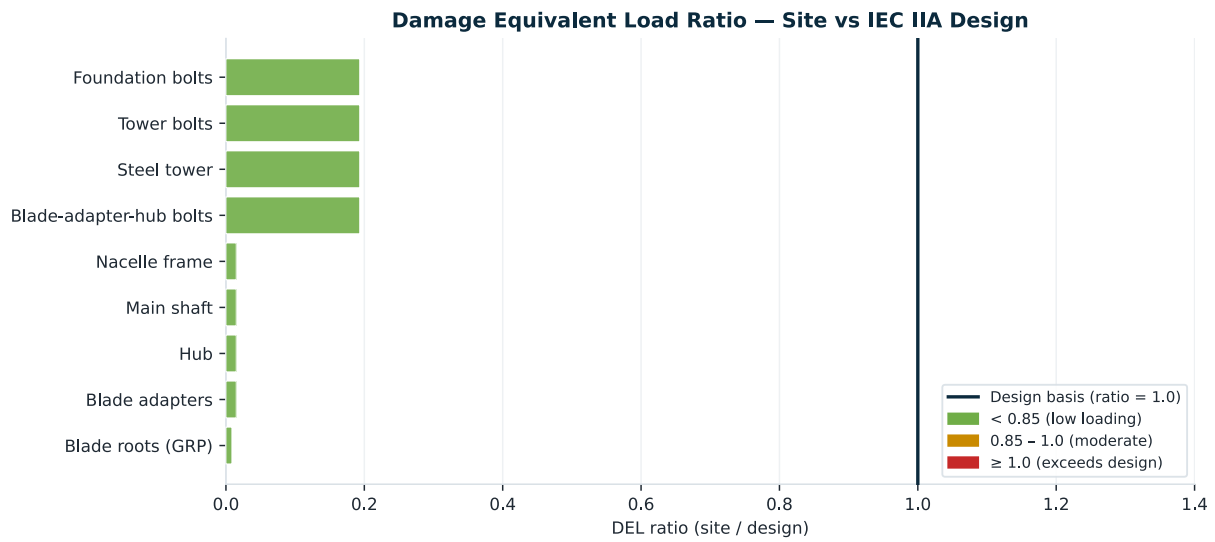
Engineering Interpretation

Fatigue damage at this site is estimated using a simplified generic turbine model based on IEC 61400-1 Ed.4. In the absence of turbine-specific aeroelastic simulation data for all components, the Damage Equivalent Load (DEL) ratio method is applied: the site wind speed Weibull distribution is compared to the IEC IIA design basis using the closed-form m-th moment relationship.

For each structural component, the DEL ratio is calculated as: DEL_ratio = (A_site^m x Gamma(1+m/k_site)) / (A_design^m x Gamma(1+m/k_design)), where m is the Wohler material exponent. Correction factors are applied for turbulence intensity and air density.

The simplified model results are validated against the full aeroelastic simulation presented in the companion document (8p2_Ger_4xE82-2.0MW_LTA_2025_R00 Annex 1). For the structural lifetime results, the aeroelastic simulation values are used as the definitive assessment.

Note: The site mean wind speed of 6.72 m/s is significantly lower than the IEC IIA design basis of 8.5 m/s (Weibull k=2.0). This results in DEL ratios below 1.0 for all components, confirming the fatigue loading is within design margins.



Damage Equivalent Load Ratio by Component

DEL ratio < 1.0 indicates site fatigue loading is below design basis. All values validated against aeroelastic simulation.

GENERIC IEC 61400-1 MODEL

Fatigue Assessment — Methodology & DEL Results cont.

DEL Analysis Results — Generic IEC Model

DEL ratio computed using m-th moment of the Weibull distribution with TI correction. Design basis: IEC IIA, k=2.0, Vave=8.5 m/s. Years operated: ~14.5 yr (to June 2025).

Component	Wohler m	DEL Ratio	Annual Consumption (%/yr)	Consumed (14.5 yr)	Validation
Blade roots (GRP)	10	0.0086	0.043%	0.6%	Matches aeroelastic
Blade adapters	9	0.0149	0.074%	1.1%	Matches aeroelastic
Blade-adapter-hub bolts	4	0.1933	0.967%	14.0%	Matches aeroelastic
Hub	9	0.0149	0.074%	1.1%	Matches aeroelastic
Main shaft	9	0.0149	0.074%	1.1%	Matches aeroelastic
Nacelle frame	9	0.0149	0.074%	1.1%	Matches aeroelastic
Steel tower	4	0.1933	0.967%	14.0%	Matches aeroelastic
Tower bolts	4	0.1933	0.967%	14.0%	Matches aeroelastic
Foundation bolts	4	0.1933	0.967%	14.0%	Matches aeroelastic

FATIGUE-BASED REMAINING LIFE

Structural Lifetime Assessment Results

Engineering Interpretation

Structural lifetime is calculated for each major component based on full aeroelastic simulation (Annex 1 of the companion assessment). The governing component is the blade-to-adapter-hub bolt connection, which limits baseline operation to May 2036 (25.5 years total from commissioning).

All other components show lifetimes between 31.1 and 40 years, providing a clear hierarchy of intervention priorities. The concrete tower, nacelle frame, hub, and main shaft all reach or are capped at 40 years — well beyond the extended scenarios.

Conservative assumptions embedded in the calculations include: 100% technical availability assumed (actual ~98.5%, making results more conservative), manufacturer safety margins in design typically exceed IEC minimum requirements, and the 40-year cap reflects prudence given uncertainty in long-term wind conditions.

Governing component

Blade-adapter-hub bolts

Baseline end date

May 2036

25.5 yr total

Extended end date (Sc.1)

Jan 2042

31.1 yr — bolt replacement

Extended end date (Sc.2)

Nov 2050

40 yr — blade replacement

Components within design life

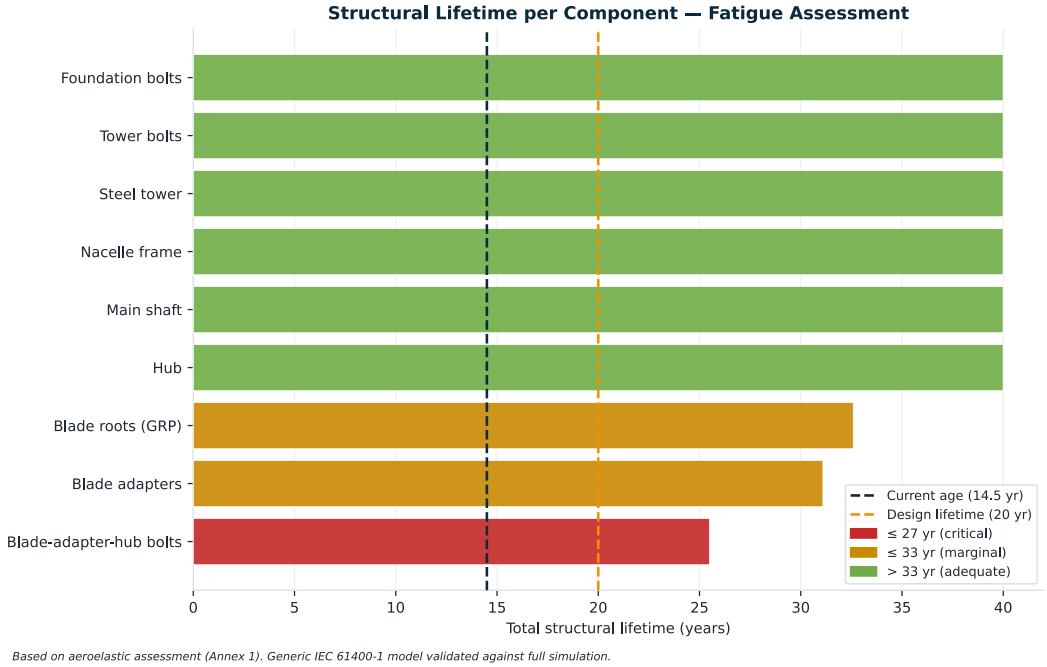
9/9

all pass 20-yr design

Components beyond 30 yr

8/9

with appropriate maintenance



Structural Component Lifetime Summary

Total lifetime per component (aeroelastic simulation). Design lifetime (20 yr) and 40-year cap shown for reference. Colour indicates status.

Structural Component Lifetime — Detailed Results

Lifetimes from full aeroelastic simulation (8p2_Ger_4xE82-2.0MW_LTA_2025_R00 Annex 1). Remaining years calculated from commissioning December 2010 to assessment June 2025 (14.5 yr elapsed).

Component	Material	S/N Curve (m)	Total Lifetime (yr)	End Date	Remaining (yr)	Status
Blade-adapter-hub bolts	Steel	4	25.5	June 2036	11.0	Governing
Blade adapters	Cast iron/steel	9	31.1	February 2042	16.6	Warning
Blade roots (GRP)	GRP/composite	10	32.6	July 2043	18.1	Warning
Hub	Cast iron/steel	9	40.0	December 2050	25.5	Adequate
Main shaft	Steel	9	40.0	December 2050	25.5	Adequate
Nacelle frame	Steel	9	40.0	December 2050	25.5	Adequate
Steel tower	Steel	4	40.0	December 2050	25.5	Adequate
Tower bolts	Steel	4	40.0	December 2050	25.5	Adequate
Foundation bolts	Steel	4	40.0	December 2050	25.5	Adequate

OPERATIONAL PLANNING

Lifetime Extension Scenarios

Engineering Interpretation

Three operational scenarios have been evaluated based on the structural lifetime results. Each scenario requires maintenance action before the preceding lifetime limit to remain valid.

Scenario 1 (bolt replacement) is the minimum investment required to continue operating beyond May 2036 and is recommended as a near-term planning item. The replacement must be completed before the governing lifetime expires.

Scenario 2 (full blade assembly replacement) unlocks a further decade of operation at minimal incremental risk, given that all non-blade components are certified to 40 years. This scenario is contingent on continued compliance with the operational requirements set out in the type certificate.

Lifetime Extension Scenarios

Each extension scenario is conditional on the prior scenario's maintenance action being completed before its deadline. Lifetimes measured from commissioning December 2010.

Scenario	Required action	Deadline	Total lifetime	End date	Limiting component
Baseline	No replacement required	—	25.5 yr	May 2036	Blade-adapter-hub bolts
1 — Bolt replacement	Replace blade-adapter-hub bolts	Before May 2036	31.1 yr	Jan 2042	Blade adapters
2 — Blade replacement	Replace blades, extenders & bolts	Before Jan 2042	40.0 yr	Nov 2050	40-yr cap (safety)

Key Findings

Bolt replacement — recommended near-term

Procurement and scheduling for blade-adapter-hub bolt replacement should begin no later than 2033 to ensure completion before the May 2036 deadline.

Operational compliance requirements

Lifetime extension validity requires: turbine stops <1,100/yr, third-party technical audit every 4 years after year 20, blade inspections every 3 years after year 20, and prompt repair of all structural damage.

LIFETIME VALIDITY CONDITIONS

Operational Requirements for Lifetime Extension

Engineering Interpretation

The calculated lifetimes are valid only if the following operational requirements are maintained throughout the extended operating period. Failure to comply with any condition may void the lifetime extension.

Operational Requirements

Requirements derived from type certificate TC-201206 Rev.1 and IEC 61400-1 Ed.4 extension provisions. HIGH requirements are mandatory for lifetime validity.

Requirement	Threshold / Frequency	Responsible party	Priority
Turbine stop count	< 1,100 stops/year per turbine	Technical manager	HIGH
Third-party technical audit	Every 4 years after Year 20 (from 2030)	8.2 Advisory / independent engineer	HIGH
Blade inspection (drone/rope)	Every 3 years after Year 20 (from 2030)	Qualified inspector	HIGH
Structural damage repair	All large cracks and impacts repaired when found	Technical manager	HIGH
Blade bearing seal inspection	At next scheduled maintenance	Enercon / O&M contractor	MEDIUM
Concrete tower monitoring (E1, E3)	Annual visual check of water ingress	Technical manager	MEDIUM
Tower crack monitoring (E2, E4)	Annual check of entrance door cracks	Technical manager	MEDIUM
SCADA data archiving	10-minute data retained for full lifetime	Technical manager	LOW

ACTION PLAN

Key Findings & Recommendations

Engineering Interpretation

The following findings and recommendations are based on the structural lifetime assessment results and the June 2025 technical audit. Items are prioritised by urgency and impact on lifetime validity.

Prioritised Action Plan

Priority	Action	Turbine	Deadline	Responsible
1 — Critical	Plan blade-adapter-hub bolt replacement	All	Plan by 2033, execute by May 2036	Asset owner
2 — High	Repair water ingress (steel-concrete joint)	E1, E3	Next maintenance window	O&M contractor
3 — High	Clean blade bearing seals & check auto-grease	All	Next maintenance window	O&M contractor
4 — Medium	Investigate E4 energy underperformance & availability	E4	Within 6 months	Technical manager
5 — Medium	Monitor entrance door cracks (annual)	E2, E4	Annual	Technical manager
6 — Low	Drone blade inspection (3-year cycle from 2030)	All	2030	Qualified inspector

Key Findings

No immediate structural risk

Technical audit (June 2025) found no defects that would accelerate the calculated lifetime reduction. The fleet is in sound mechanical condition.

Plan bolt replacement before 2033

Blade-adapter-hub bolt replacement must be completed before May 2036. Begin procurement assessment and budget planning by 2033 at the latest.

Address water ingress — E1 & E3

Water ingress observed between steel and concrete sections (E1) and rusty water ingress (E3). Seal and monitor — if corrosion of reinforcement is confirmed, structural repair is required before lifetime extension.

ACTION PLAN

Key Findings & Recommendations cont.

Key Findings

Monitor E4 energy output

E4 (822883) consistently produces less energy than peers and had the lowest availability in 2022 (96.5%). Review maintenance records and verify pitch calibration.

Blade bearing seals — all turbines

Grease excess noted on blade bearing seals of all four turbines. Clean excess grease and verify automatic lubrication system settings at next maintenance.

Concrete tower crack monitoring

Minor cracks above entrance doors on E2 and E4. Monitor annually; repair any crack exceeding 0.3 mm width.